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Cohort profile: baseline characteristics and design of the McMaster Monitoring My Mobility (MacM3) study – a prospective digital mobility cohort of community-dwelling older Canadians from Southern Ontario

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ABSTRACT

Purpose The McMaster Monitoring My Mobility (MacM3) study aims to understand trajectories of mobility decline in later life using multisensor wearable technology. To our knowledge, MacM3 is the first major cohort to combine accelerometry and a Global Positioning System (GPS) to track real-world mobility in community-dwelling older adults.

Participants Between May 2022 and May 2024, MacM3 recruited 1555 community-dwelling older adults (mean age 73.9 years, SD=5.5) from Hamilton and Toronto, Ontario. Of the cohort, 68.4% were female, 62.4% married/partnered, 75.3% had post-secondary education and 62.9% had≥3 comorbidities. Most were Canadian born (69.4%) and white/Caucasian (88.0%), with greater ethnocultural diversity observed at the Toronto site.

Findings to date At baseline, 56.7% of participants reported no mobility limitations, 15.9% had preclinical limitations and 27.4% had minor mobility limitations. Mean gait speed for the total sample was 1.23 m/s, with a mean Timed Up and Go time of 9.4 s and a 5× sit-to-stand time of 13.0 s. A total of 1301 participants had valid wrist-worn device data, and 1008 participants who agreed to wear the thigh-worn device had valid data (≥7 days with ≥10 hours of wear per day). Step count data (n=1008) revealed a mean of 8437 steps per day (SD=2943), with 5073 steps in the lowest quartile and 12 303 steps in the highest.

Future plans Ongoing work aims to develop predictive models of mobility decline by integrating wearable, clinical and environmental data. Pipeline enhancements will enable GPS/inertial measurement unit fusion to explore mobility-environment interactions and support ageing-in-place tools.

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ McMaster Monitoring My Mobility (MacM3) is a prospective cohort study tracking mobility in community-dwelling adults aged 65+ from Hamilton and Toronto, recruited using diverse, community-engaged strategies.
⇒ Participants wear a wrist-based TicWatch and an optional thigh-worn ActiGraph or Axivity device for 10 days every 4 months, collecting high-resolution accelerometry and GNSS (Global Navigation Satellite System) data to assess activity, posture and mobility patterns.
⇒ Digital mobility measures include activity intensity, postural behaviour, walking patterns and geospatial mobility, derived from synchronised wrist and thigh sensors.
⇒ Health, psychosocial and demographic data are collected using validated tools administered during quarterly telephone interviews and yearly in-person visits with strong participant retention maintained over 2 years of follow-up to date.
⇒ Stakeholder engagement informed the study design and led to inclusive, participant-centred protocols that enhance recruitment, retention and generalisability of the MacM3 cohort.

INTRODUCTION

In the 2015 World Report on Ageing and Health, the WHO defines healthy ageing as the 'process of developing and maintaining the functional ability that enables well-being in older age'.[1] Central to this concept is mobility—the ability to move, walk and use transportation within our homes and communities.[2] Mobility is a cornerstone of healthy ageing, reported to be of utmost importance to older people as they age, and has been labelled by many ageing experts as the

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'6th vital sign'.[3–5] With population ageing, the number of older adults living with mobility limitations will rise dramatically over the next four decades. In Canada, one in two older adults report reduced mobility compared with the previous decade, with 24–33% facing challenges in conducting daily activities that require physical capabilities, such as climbing stairs and walking.[6–8] These limitations can signal early neurological, musculoskeletal or cardiorespiratory health changes and predict adverse outcomes like falls, hospitalisation, institutionalisation and mortality.[9–12]

Despite ongoing efforts to understand predictors and trajectories of mobility decline in older adulthood, knowledge gaps remain, particularly in understanding the varied trajectories of mobility changes across different age groups and functional levels.[13–18] Previous studies have predominantly focused on older populations (age 70+) already experiencing mobility limitations or relied on self-reported measures of function, which may not accurately capture day-to-day mobility challenges.[13 14 16–18] As such, our understanding of the actual level and types of mobility undertaken by individuals in their daily lives is limited.[19] By expanding our research to include a broader age range and incorporating device-based measures of real-world mobility, we can effectively identify early signs of decline and inform interventions to delay or prevent the onset of disability.

Advancements in wearable technology have revolutionised our ability to continuously monitor unsupervised mobility in individuals' daily lives, both at home and in the community ('free-living').[20–22] This innovation has generated substantial interest and great potential to use wearables for prevention, health management and personalised care, especially among older adults who increasingly recognise the value of self-monitoring their mobility as part of their health regimen.[23 24] Our consultations with Canadian stakeholders (ie, older adults, caregivers and industry and community partners) confirmed that older adults have a strong interest in using such technology to monitor their mobility and enhance their health. Given that mobility data obtained using wearable technology (ie, digital mobility measures) reflect real-world functioning, these measures hold exceptional potential for monitoring and predicting health outcomes and promoting ageing-in-place.[25 26]

In 2022, we established the McMaster Monitoring My Mobility (MacM3) research platform, with the primary objective of characterising later-life trajectories of early mobility decline through longitudinal, multisensor-based data collected from a large cohort of older adults exhibiting differing levels of mobility at baseline. This pioneering initiative represents the first major longitudinal study to evaluate real-world mobility with frequent repeated measures in community-dwelling older adults using a wearable system that integrates Global Positioning Systems (GPS) and accelerometry.[27] Our methodology centres on a wrist-worn device that synchronises time-based accelerometry with GPS data, complemented by a thigh-worn accelerometer. The unique combination of accelerometry and geolocation data provides a holistic view of mobility, capturing interactions between individuals and their environments. Such a comprehensive approach enhances our capacity to detect early signs of mobility decline and develop strategies to mitigate mobility-related adverse health impacts in older adulthood.

COHORT DESCRIPTION

Study design and recruitment

MacM3 is a prospective cohort study with 24 months of follow-up initiated in May 2022 at the Hamilton, Ontario site and expanded to the Dixon Hall social services site in Toronto in January 2023. The study targeted individuals aged 65 years and older residing within a 50 km radius of McMaster University in Hamilton or Dixon Hall in Toronto. All participants provided verbal informed consent. Eligibility criteria included proficiency in English or either English or Mandarin (at Dixon Hall) and classification into one of three mobility groups based on the concept of preclinical mobility limitation[28] measured by the Preclinical Mobility Limitation (PCML) [29]scale: (1) no mobility limitation, (2) preclinical mobility limitation or (3) minor mobility limitation. The PCML classification is based on self-reported difficulty with three mobility tasks: walking 0.5 km, walking 2 km and climbing one flight of stairs. For each task, participants are asked whether they are: (1) able to manage without difficulty, (2) able to manage with some difficulty, (3) able to manage with a great deal of difficulty, (4) able to manage only with the help of another person or (5) unable to manage even with help. Participants who report being able to manage any task without difficulty (response 1) are further asked about task modification using a set of follow-up questions. These include: 'I am able to walk 0.5 km the same way I always have', 'I need to rest in the middle of the walk', 'I walk 0.5 km less often than I used to', 'It takes me longer to walk 0.5 km than it used to' and 'I am more tired after walking 0.5 km than I used to be'. Participants who report no task modification—that is, they perform all tasks the same way as before—are classified as having *no mobility limitation*. Those who report modification of any task (eg, reduced frequency, slower performance, fatigue or need for rest) are classified as having a *PCML*. Individuals who report being able to manage a task with some difficulty (response 2) are classified as having a *minor mobility limitation*, while those who report great difficulty, needing help or being unable to complete a task (responses 3, 4 or 5) are classified as having a *major mobility limitation*. While individuals with major mobility limitations were excluded at baseline, they could be reclassified into this category during follow-up assessments. Additional exclusion criteria included inability to communicate in required languages, significant visual or hearing impairments that could affect safety during assessments and plans to relocate outside the study area within 2 years.

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We engaged in a multipronged recruitment strategy aimed at enhancing the diversity of participants in the study (in ethnicity, race, income level, culture, ability and region), and ensuring the data reflects diverse participant experiences. Our recruitment strategy employed a combination of community-based approaches, including word of mouth, advertisements in local newspapers and online platforms. Targeted mail campaigns were conducted using Canada Post Snap Admail, focusing on postal routes and codes that matched the study's demographic in lower-income areas, and advertisements were placed on relevant public transportation routes. Our recruitment strategy was designed to reach adults aged 65 and older—including males, females, individuals of any gender identity and people living in both higher- and lower-income areas—to ensure a diverse and inclusive sample.

Our second recruitment site, Dixon Hall, serves a large, diverse population, including many individuals who are under-represented in research, particularly those facing social challenges such as poverty, isolation and health inequities. Our partnership with Dixon Hall, facilitated by the McMaster Institute for Research on Aging (MIRA) | Dixon Hall Centre, allows us to engage with a population that is often excluded from research studies, ensuring that the data collected are more generalisable to the broader community of older adults. In order to build trust and access, we embedded researchers at the Dixon Hall site, recruiting participants active in social and recreation groups via a snowball method, where participants invited peers to participate. While the Hamilton participants tend to be more homogenous (Caucasian and middle-income), Dixon Hall participants had more diversity in race, income level, ethnicity and primary spoken language (see table 1). Additionally, a small subset of participants (n=148) was recruited from the MIRA-iGeN (MIRA Intergenerational Study) platform, which includes 5000 individuals prerecruited by Canadian Viewpoint through phone, online and mail-out methods.

Our sample size was designed to support MacM3's main objectives: to compare mobility patterns among older adults with varying levels of early mobility limitation; identify distinct mobility trajectories and their characteristics; assess the predictive value of digital mobility measures for adverse outcomes; and model risk factors for the onset and progression of mobility decline in older Canadians. To allow for trajectory analyses and to account for potential seasonal changes in activity patterns, we incorporated frequent repeated follow-ups (every 4 months).

Based on our original objectives, our target sample size was 1500 participants, grounded in both feasibility and predictive validity considerations. Prior studies examining the accuracy of mobility measures for predicting health outcomes have reported area under the curve (AUC) values ranging from 0.6 to 0.8.[30 31] To ensure adequate power (80%) to detect a significant deviation from the null hypothesis (AUC=0.5; two-sided test at $p<0.05$), we based our calculations on the most conservative expected AUC of 0.6. Based on conservative expected incidence rates for key outcomes: at least one fall (20% over 1 year), visit to the emergency room (12% over 1 year), overnight hospital stays (8% over 2 years) and progression to major mobility disability (10% over 2 years), a sample of approximately 740 participants would be needed to detect progression to major mobility limitation over 2 years, and around 634 participants to detect emergency room visits over the same period. Accounting for an anticipated dropout rate of 20% and the potential for missing device-based data of up to 20%, the required sample increases to approximately 1156 for progression to major mobility disability and 991 for emergency room visits. This sample size is also sufficient for detecting small and clinically relevant differences between the three mobility level groups based on the PCML for step count (400–600 steps) and for trips per week (4–6 trips). Our flowchart (figure 1) updated on 23 September 2025 highlights that we completed our 12-month follow-up with 1378 active participants.

Study procedures

Participants in MacM3 are asked to wear a consumer grade smartwatch with Global Navigation Satellite System (GNSS) and activity monitoring capabilities, the TicWatch Pro 3 Ultra GPS[27] on the wrist for 10 days at baseline and at 4-month intervals for a total of seven time points of data collection over 24 months. At the time of consent, participants were also asked if they would be willing to wear a second (optional) activity monitoring device, the ActiGraph wGT3X-BT[27 32] or the Axivity X3,[33] on the thigh concurrently with the wrist-worn device. At baseline, 12 months and 24 months, participants complete a more in-depth series of questionnaires about their health and mobility through a combination of telephone interviews and online surveys as well as an in-person physical assessment. In between the yearly assessments, devices are mailed to participants' homes every 4 months and then returned to the study site via prepaid shipping. To monitor health outcomes (eg, new diagnoses, healthcare utilisation, injuries, falls), we use quarterly phone calls and monthly mail-in fall diaries.[34] Figure 2 provides an overview of the MacM3 study design.

Study measures

Our data collection encompasses all three 'tenses' of mobility/function in line with the model first proposed by Glass[35] and outlined more recently by members of our team based on the World Report on Ageing and Health[36]: enacted (ie, actual mobility in the community measured via wearables), hypothetical (ie, perceived ability measured via self-reported questionnaires) and experimental (ie, locomotor capacity for mobility measured via in-person physical assessments). Self-reported measures of mobility provide information on what patients 'can do'; performance-based testing provides information on what patients 'could do'; and importantly, the sensor data on enacted/actual mobility provide information on what patients actually 'do' in the real world.[36]

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Table 1 Outcome measures and time points of the MacM3 cohort study

Model of assessment	Outcomes		Follow-ups						
	Measures		Base-line	4M	8M	12M	16M	20M	24M
Free-living mobility protocol	Wearables	Accelerometry and GPS data	✓	✓	✓	✓	✓	✓	✓
Phone call assessments	Mobility/function	Late Life Function and Disability Instrument	✓			✓			✓
		Life-space assessment	✓			✓			✓
		Mobility screening questions	✓	✓	✓	✓	✓	✓	✓
		Global Rating Change scale	✓	✓	✓	✓	✓	✓	✓
	Driving mobility	Transportation mobility questions	✓			✓			✓
		Driving Comfort Scale	✓			✓			✓
		Perceiving Driving Abilities	✓			✓			✓
In-person assessments	Performance-based tests and other measures	Mobility Assessment Tool; TUG (regular/dual-task) Repeated Sit-to-Stand test; gait speed 3 m (usual /fast pace); single leg stance; stair climbing power test; 400-metre walk test; visual acuity test; lung function; Clinical Frailty Scale and grip strength	✓			✓			✓
	Comprehensive health and well-being measures								
Phone call assessments	Demographics, socioeconomic and health status	Age, country of birth, race and ethnicity, education	✓						
		Height, weight, sex and gender, marital status, employment situation, income, housing and living situation, self-reported chronic conditions, smoking, alcohol use, care giving and care receiving, medication, nutritional supplements and cannabis use, pain, balance, fall history and fear of falling, neighbourhood environment, technology	✓			✓			✓
		Healthcare utilisation, new diagnoses, general health, mental health, car crashes, mobility aids, falls and near falls information and injuries	✓	✓	✓	✓	✓	✓	✓
	COVID-19	COVID-19 and vaccination status	✓	✓	✓	✓	✓	✓	✓
	Cognition	Montreal Cognitive Assessment (telephone version)	✓			✓			✓

Continued

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Table 1 Continued

Model of assessment	Outcomes		Follow-ups						
	Measures		Base-line	4M	8M	12M	16M	20M	24M
	Physical activity	Physical Activity Scale for the Elderly	✓			✓			✓
	Social support	*Medical Outcome Study Social Support Survey	✓			✓			✓
	Depression	*The Centre for Epidemiological Studies Depression Scale	✓			✓			✓
	Hearing	*Hearing Handicap Inventory for the Elderly Screening version	✓			✓			✓
	Nutrition	*Risk Evaluation for Eating and Nutrition (SCREEN-8)	✓			✓			✓
	Quality of life	*EuroQol-5D-5L	✓			✓			✓
	Anxiety	*General Anxiety Disorder	✓			✓			✓
	Loneliness	*20-item Loneliness scale (UCLA)	✓			✓			✓
	Social network	*Lubben Social Network Scale	✓			✓			✓
	Pain	*Brief Pain Inventory				✓			✓
Monthly diaries	Falls	Occurrence of falls; injurious falls	Monthly						

*Option to complete during the in-person physical assessment or at home in a survey format.
GPS, Global Positioning System; M, month; MacM3, McMaster Monitoring My Mobility; SCREEN-8, Seniors in the Community: Risk Evaluation for Eating and Nutrition; TUG, Timed Up and Go; UCLA, University of California, Los Angeles.

Mobility measures

Enacted mobility

The MacM3 study employs a multisensor, multinodal approach for studying enacted and free-living mobility, integrating time-synchronised accelerometry with GPS in a wrist-worn device alongside a second thigh-worn accelerometer.[27] Device selections and wear locations for the current cohort were made after extensive laboratory-based testing, consultation with our older adult Stakeholder Advisory Committee and pilot testing to determine their reliability and technical validity.[27]

Participants are asked to wear a TicWatch Pro 3 Ultra GPS on their non-dominant wrist and either the ActiGraph GT3X-BT or an Axivity AX3 on the thigh for 10 consecutive days during waking hours at baseline and every 4 months throughout the study. The *wrist-worn device* (TicWatch by Mobvoi) is a smartwatch that uses the Wear OS operating system by Google and has a built-in tri-axial accelerometer, gyroscope, heart rate sensor and multi-GNSS (GPS, Global Navigation Satellite System (GLONASS), Galileo, Quasi-Zenith Satellite System (QZSS) and BeiDou) for logging position coordinates and time. As the watch was intended to be used in our study as a standalone data logging device without a cellphone, GNSS data are collected at different intervals depending on the predicted activity state of the device using Google's Activity Recognition Application Programming Interface (API). To improve battery life, we configured the device to collect location coordinates every 5 s when the watch is in an active motion state (eg, walking, in a vehicle) with good satellite reception, and less frequently when the device is still or there is poor satellite reception.[27] We validated the TicWatch against a gold-standard standalone GPS device (Qstarz GPS) and validated its accelerometer output against the thigh-worn ActiGraph device. In this validation study, we found that the TicWatch models were valid devices for capturing both GPS and raw accelerometer data, supporting their use as reliable tools for assessing real-life mobility in older adults.[27]

The second device is *a thigh-worn accelerometer* (either the ActiGraph wGT3X-BT or Axivity AX3) secured to the upper thigh using an adhesive patch (Hypafix). The ActiGraph wGT3X-BT[37] (ActiGraph, Pensacola, Florida, USA) and Axivity AX3[33] (Axivity, UK) are commonly used in epidemiological research for measuring physical activity[38–40] and body posture, with thigh placement considered the gold standard for accurately distinguishing between sitting, standing and walking.[27 41–44] Both devices have built-in triaxial accelerometers for recording high-resolution raw acceleration data and long-lasting batteries that eliminate the need for recharging during the data collection period. In our validation study, the ActiGraph

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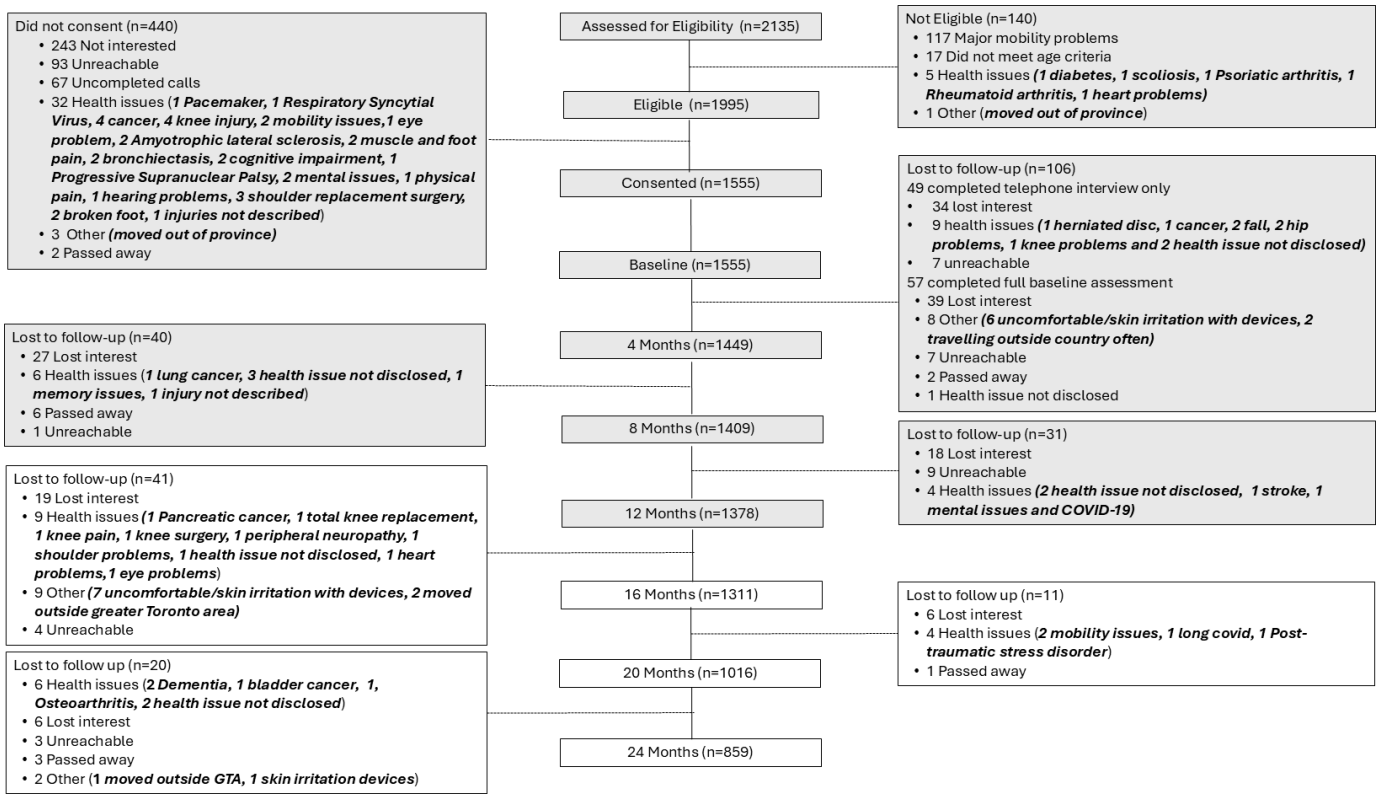


Figure 1 Flow of study participants and timeline of data collection in the MacM3 study, from recruitment through September 2025. Grey areas indicate completed follow-up periods. MacM3, McMaster Monitoring My Mobility; GTA, Greater Toronto Area.

demonstrated high accuracy in detecting body posture. Full details of this validation study have been published elsewhere.[27]

A custom data collection application, Ivy, was designed by our team exclusively for the TicWatch to optimise data quality and battery life by collecting data from onboard sensors, including the accelerometer, gyroscope, geolocation and heart rate.[27] A companion app, Clover, supports Ivy on Windows by automating the setup of both the TicWatch and the Axivity device, streamlining data downloads and device preparation. The ActiGraph device, on the other hand, is initialised using the ActiLife proprietary software. Additionally, Clover integrates information on the date, time and completion status

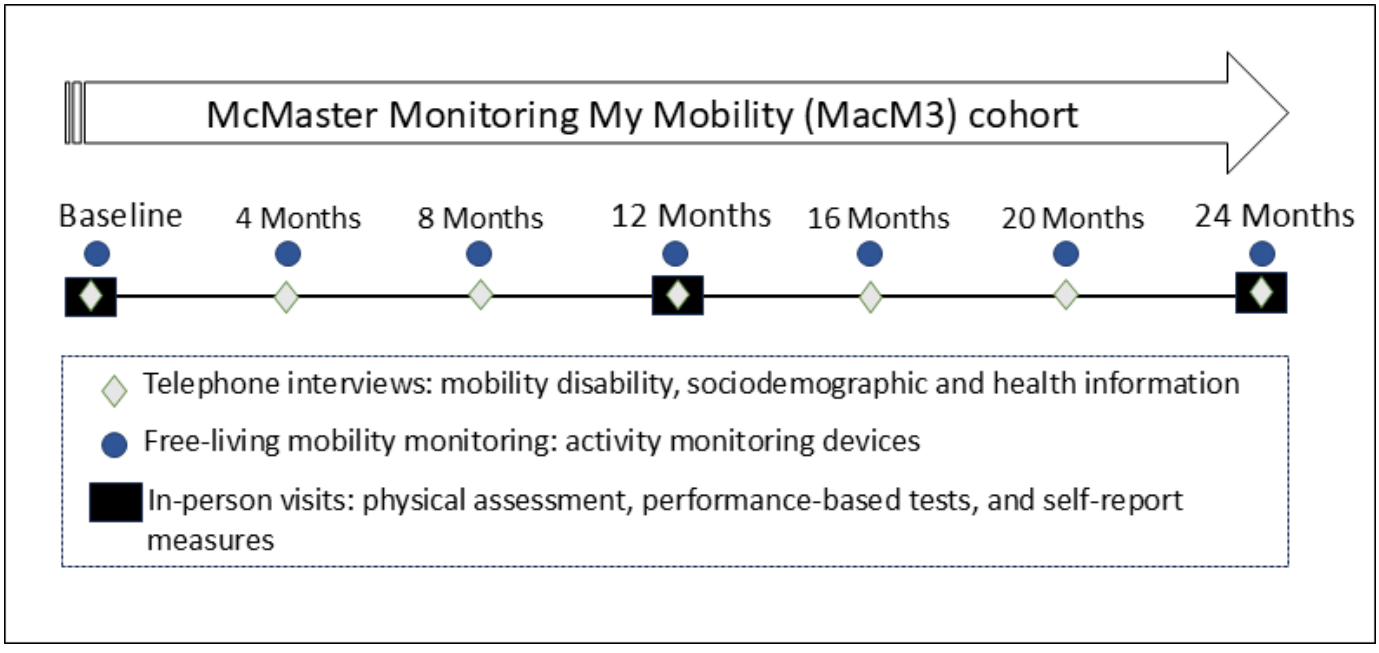


Figure 2 MacM3 study overview.

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of data downloads with our Research Electronic Data Capture (REDCap) data collection system for all devices, including the ActiGraph.

The primary digitally-derived mobility measures of interest for MacM3 can be classified into five core domains: *activity* (amount of time spent in sedentary, light, moderate and vigorous physical activity),[45 46] *volume-related features of walking* (steps/day, walking bout lengths and duration), *spatiotemporal features of walking* (gait cadence, variability, velocity and stride width during long bouts),[47] *postural* (time spent lying down, sitting, standing and frequency of transitions)[42 48] and *geospatial features* (trip frequency, trip duration, trip mode, maximum distance from home and minimum convex hull/life space area).[49 50] The derivation of these digital mobility measures has been described in previous technical validation papers by our team and others.[27 51–53] Notably, with integrated GNSS and accelerometry data, we are able to uniquely distinguish and characterise indoor and outdoor mobility (bouts, trips) and patterns of mobility outside of the home.

All measures are processed using the NiMBaLWear[51] pipeline which is a modular, open-source wearable sensor analytical framework designed to accommodate a wide range of sensors, signal types and body wear locations. NiMBalWear was developed to address the limitations of device-specific processing by providing a standardised, flexible approach for analysing raw accelerometry data across different platforms. This framework unifies the measurement and analysis of multiple health-related domains—such as mobility, posture and physical activity—by optimising data extraction from individual sensors and enabling data fusion across multiple sensor types and locations when applicable. By focusing on raw signal processing, NiMBaLWear ensures device-agnostic harmonisation during the feature extraction stage, allowing for consistent, comparable metrics regardless of brand or configuration. This is particularly important for longitudinal studies or multisite research using different devices, as it supports reproducible and scalable analysis without reliance on proprietary algorithms or cut-points. Currently, the NiMBaLWear platform also includes a Mobility module focused on inertial measurement unit (IMU-based) outcomes, which is being expanded to incorporate GNSS+analytics.

NiMBaLWear-based processing of thigh-worn and wrist-worn accelerometry data

Step counts and walking bouts are derived from the thigh-worn accelerometer. NiMBaLWear applies wavelet-based time-frequency analysis to the triaxial acceleration signal to detect unilateral steps.[54 55] Step-related events are identified based on the frequency content of the acceleration profile, capturing both the swing and stance phases of walking. Foot contact is used to define the timing of each step following a detected swing phase. Consecutive steps are then grouped into walking bouts according to predefined timing criteria (eg, a minimum number of steps within a maximum interval). This method offers greater precision than traditional threshold-based techniques and has demonstrated improved accuracy in detecting steps and walking patterns, particularly in older adults. The step count data reported in this manuscript align with findings from other large cohort studies[56 57] that applied machine learning methods to estimate step count from wrist-worn devices and observed similar trends—namely, a decline in step counts with advancing age and increased mobility limitations.

Postural behaviour is derived from the same thigh-worn triaxial accelerometer using algorithms based on the orientation of the device relative to gravity. Specifically, posture was classified as either standing or sitting/lying based on the acceleration due to gravity recorded along each axis of the accelerometer, assuming consistent device placement on the thigh. Postural transitions between standing and sitting/lying were detected by identifying periods of sustained change in thigh orientation that met predefined criteria. Once a transition was detected, all subsequent time points were classified according to the last identified posture. This method allows us to determine body posture continuously throughout the wear period using a single device, with classification based solely on orientation and transition patterns.

Activity intensity, including sedentary behaviour, light physical activity and moderate-to-vigorous physical activity, is derived from the wrist device. We applied epoch-independent cut-points validated by Fraysse *et al*,[58] which classify activity based on raw acceleration (in mg). Specifically, thresholds of 42.5 mg (non-dominant wrist) and 62.5 mg (dominant wrist) were used for light activity, while 98.0 mg (non-dominant) and 92.5 mg (dominant) defined moderate activity. These cut-points have been applied in other ageing cohorts and support comparability across studies examining accelerometry-derived activity patterns.

Hypothetical mobility

Perceived mobility is assessed through various self-report instruments developed to capture different mobility domains among older adults.[59 60] The Late-Life Function and Disability Instrument[61 62] evaluates both physical function mobility and participation in valued life roles, encompassing tasks from basic physical actions to community activities. The Mobility Assessment Tool – short form[63] is a 10-item computer-administered measure of perceived mobility using video technology. The Life-Space Assessment[64] measures life-space mobility—the distance, frequency and independence perceived by individuals as they move through their communities. Transportation mobility is evaluated through questions from the Canadian Longitudinal Study on Ageing[65] Transportation, Mobility, Migration[66] module, while driving behaviour is assessed using the Driving Questionnaire, Driving Comfort Scale[67 68] as well as selected questions from the Perceived Driving Abilities Scale.[67]

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Experimental mobility

Participants undergo in-person physical assessments at either the Hamilton or Toronto location at baseline, 12 months and 24 months follow-up. These assessments include commonly used and well-validated performance-based tests of mobility and physical function, namely the Timed Up and Go (TUG)[69] usual and dual-task tests, the repeated sit-to-stand test (5× STS), tests of gait speed[70] (both usual and fast pace) and single-leg balance.[71] The 400-metre walk test[72 73] evaluates aerobic capacity, while muscle power is estimated with the stair-climbing power test.[74 75] Other physical measures collected at the in-person assessment include the Clinical Frailty Scale,[76] visual acuity test,[77] spirometry[78] for pulmonary function evaluation and handgrip strength[79] assessment. Participants are permitted to use their mobility aids during the TUG, gait speed, stair-climbing and 400metre walk tests. Use of mobility aids was documented and recorded in the study database for future analyses.

Demographic and health measures

As part of the yearly in-person assessments, participants also complete a series of questionnaires related to their basic health and the possible physical, psychosocial, cognitive, financial and environmental contributors to mobility. This encompasses demographics, such as socio-economic status and lifestyle factors, as well as health-related behaviours. We assess physical activity through the Physical Activity Scale for the Elderly,[80] nutrition risk using the Seniors in the Community: Risk Evaluation for Eating and Nutrition,[81] pain using the Brief Pain Inventory-Short Form[82] and health status with the EuroQol-5D-5L.[83] Additionally, cognitive[84] and hearing[85] function, social support[86–88] and mental health[89 90] are assessed through validated tools. These measures are captured using a combination of telephone interviews and online surveys at baseline, 12-month and 24-month follow-ups.

To monitor health and mobility outcomes longitudinally, participants are contacted at 4-month intervals from baseline until the conclusion of the study. During these calls, the PCML[91] scale is re-administered to track changes over time and to identify the development of mobility disability. Participants also rate their perceived changes in mobility using a Global Rating of Change[92] scale and answer questions about healthcare utilisation, including hospitalisations, emergency room visits, primary care visits, rehabilitation service use, motor vehicle collisions, new diagnoses, living situation updates and general and mental health status.

Falls are measured prospectively using monthly fall diary calendars, which are provided to participants along with pre-addressed and pre-stamped postcards. This method of data collection for falls is regarded as the gold standard by international expert consensus.[34] As part of the data quality checks, the research team calls participants to rectify missing data and confirm reported falls, ensuring they fit the definition of a fall[34 93] and documenting circumstances. Participants who report falls are requested to complete a standardised case report form via telephone to collect detailed information, with fall-related injuries categorised as serious, moderate or minor based on symptoms and healthcare use. Participants are encouraged to report injurious falls promptly for accurate documentation. Table 1 provides an overview of the measures and corresponding assessment time points.

Patient and public involvement

Older adult stakeholders (n=7) were actively engaged throughout the development of the MacM3 cohort study. They provided valuable feedback on various aspects of the project, including the study concept (interest and perceived value of mobility monitoring) and data collection methods (ie, selection of the wearable device and wear location). While older adult stakeholders were not directly involved in defining the research questions or selecting outcome measures, their feedback informed the perceived value, feasibility and burden of study procedures.

We plan to involve both stakeholder and community advisory groups in the interpretation and dissemination of study findings, including selecting relevant information and appropriate formats for sharing results with participants and wider communities. We will continue to consult with our adult partners regarding the interpretation of study findings, and strategies for translating research outcomes into practical applications. This ongoing engagement helps ensure the research remains relevant to the needs and concerns of the community it seeks to benefit. To maintain this strong connection, the project plans to meet with stakeholders at least once a year, offering a forum for continued dialogue and input.

As part of our partnership with the Dixon Hall Centre in Toronto, we also consulted extensively with older adults and social service providers in the community to ensure our data collection methods were as inclusive as possible. A community advisory group was assembled (11 members) who continue to provide input on all stages of our research at the Centre which has resulted in key modifications to our protocol to facilitate participant recruitment and retention including: options for returning the device in person, Mandarin speaking assessors, drop-in hours for troubleshooting, gift cards for returning devices on time, option to complete online survey in person and distributing flyers that outlined our commitment to address the under-representation of diverse older adults in research—while also emphasising safety and security. This comprehensive approach to stakeholder involvement ensures that the MacM3 cohort reflects a wide range of experiences and perspectives, strengthening the overall impact and generalisability of the research.

RESULTS

Participants characteristics

We aimed to enrol a total of 1500 participants, and as of May 2024, recruitment was completed with 1210

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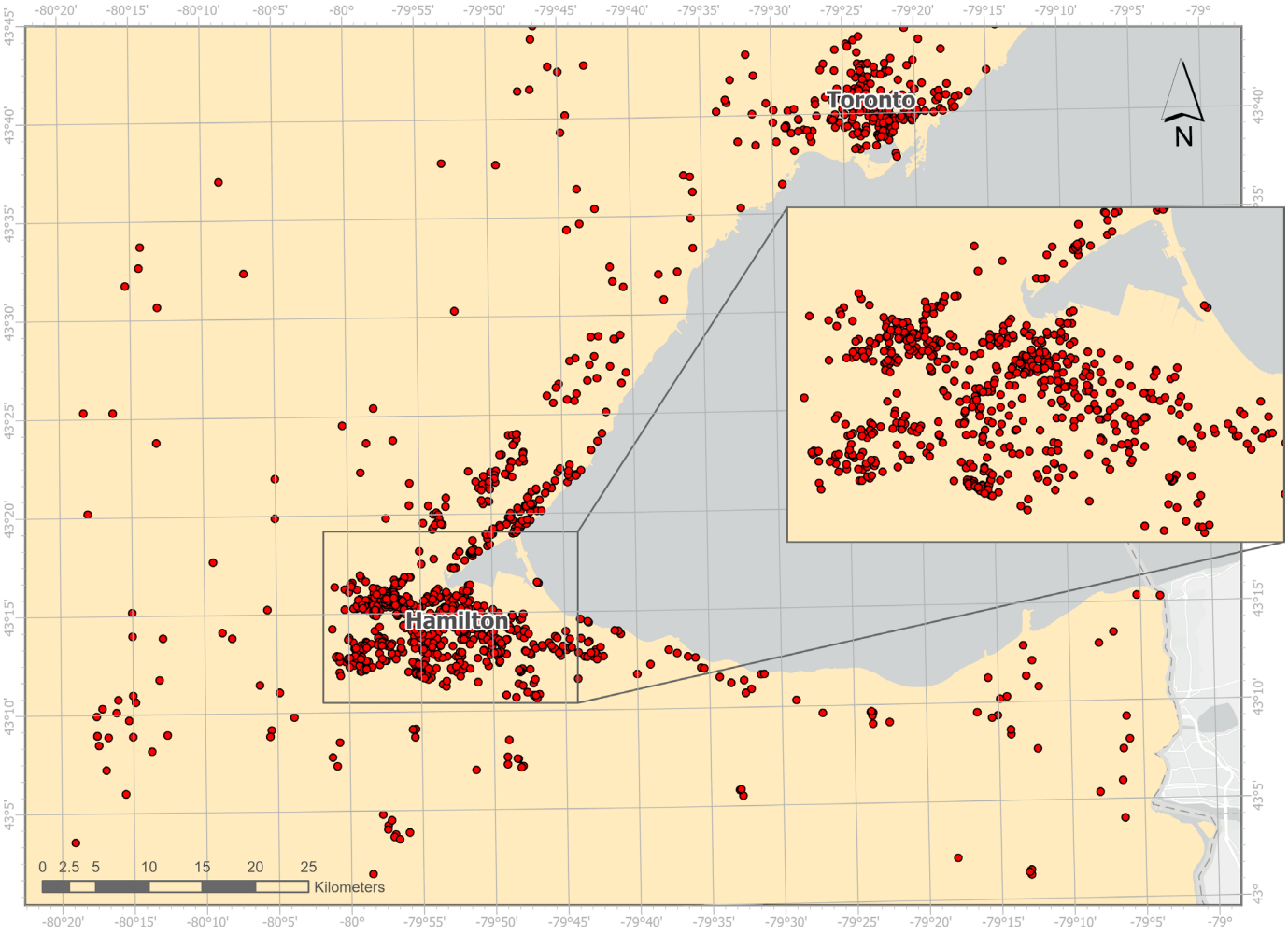


Figure 3 Geographical distribution of MacM3 participants. MacM3, McMaster Monitoring My Mobility.

participants enrolled at the Hamilton site and 345 at the Dixon Hall site, for a total of 1555 participants. Figure 3 displays the geographical distribution of participant addresses, demonstrating a concentration of recruitment in Hamilton and Toronto, Ontario, aligning with the study's targeted catchment area. The inset highlights the dense and well-distributed recruitment pattern within Hamilton, indicating strong geographical coverage across the city. Based on Statistics Canada's Population Centre and Rural Area Classification, most participants (n=1416) reside in large population centres. A smaller number live in medium (n=5) and small (n=57) population centres, and 76 participants were recruited from rural areas. One participant's address was missing and could not be classified.

Figure 1 illustrates the flow of study participants from recruitment through each follow-up period, highlighting participant retention and attrition trends over time. A total of 2135 individuals expressed interest in the MacM3 study. Of these, 140 were deemed ineligible—primarily due to major mobility limitations (n=117), based on our PCML screening survey. Additionally, 440 individuals declined to participate. As a result, 1555 participants consented to the MacM3 study. Baseline assessments were completed for all enrolled participants, along with 4-month, 8-month and 12-month follow-ups, with 1378 participants completing the 12-month assessment. The study continues to demonstrate good retention, with an overall attrition rate of just 16.0%. The highest dropout rate occurred at the beginning of the study, between baseline and 4 months, where many participants (n=57, 3.7%) dropped out after completing all components of the baseline assessment. The most common reasons for dropout included participants losing interest, becoming unreachable, device discomfort or experiencing health-related challenges. Mortality has been reported in 12 participants (0.7%) during follow-up to date. As of the latest follow-up, 859 participants have successfully completed their 24-month follow-up; data collection is still ongoing. The study's overall high retention rate underscores the feasibility of long-term data collection using wearable devices and highlights the effectiveness of our engagement strategies in sustaining participant involvement (figure 1).

Table 2 presents a comprehensive overview of the sociodemographic, lifestyle and health characteristics of the MacM3 participants at baseline. Among the total cohort (n=1555), 68.4% were female and the mean age at baseline was 73.9 years (SD=5.5). A majority of participants

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Table 2 Selected baseline characteristics of the MacM3 participants

			Hamilton site		Dixon Hall site	
		Total	Males n (%)	Females n (%)	Males n (%)	Females n (%)
Sample size, n (%)		1555	408 (33.7)	802 (66.3)	83 (24.0)	262 (76.0)
Age (years), mean (SD)		73.9 (5.5)	74.4 (5.7)	73.5 (5.4)	74.8 (5.8)	73.8 (5.5)
BMI (kg/m^2), mean (SD)		26.3 (4.8)	27.0 (4.2)	26.4 (4.9)	25.8 (5.9)	25.2 (5.0)
Country of birth, n (%)	Born in Canada	1079 (69.4)	296 (72.6)	621 (77.4)	35 (42.2)	127 (48.5)
Ethnicity (participants could select multiple ethnicities)	White/Caucasian	1369 (88.0)	388 (95.1)	770 (96.0)	48 (57.8)	163 (62.2)
	Chinese	108 (7.0)	1 (0.2)	4 (0.5)	32 (38.6)	71 (27.1)
	Other (black, Japanese, South Asian, Latin American, North American Indian, Metis, Arab, Korean, other)	88 (5.7)	25 (6.1)	31 (3.9)	2 (2.4)	30 (11.4)
Marital status, n (%)	Married/living with a partner	970 (62.4)	337 (82.6)	457 (57.0)	67 (80.8)	109 (41.6)
	Widowed	229 (14.7)	22 (5.4)	158 (19.7)	4 (4.8)	45 (17.2)
	Divorced	174 (11.2)	15 (3.7)	110 (13.8)	8 (9.6)	41 (15.6)
	Separated	36 (2.3)	9 (2.2)	16 (2.0)	0 (0.0)	11 (4.2)
	Single	145 (9.3)	25 (6.1)	60 (7.5)	4 (4.8)	56 (21.4)
Education, n (%)	Secondary school incomplete	54 (3.4)	12 (2.9)	26 (3.3)	2 (2.4)	14 (5.4)
	Secondary school completed	158 (10.2)	53 (13.0)	74 (9.2)	6 (7.2)	25 (9.5)
	Some post-secondary education	172 (11.1)	44 (10.8)	103 (12.8)	7 (8.5)	18 (6.9)
	Post-secondary diploma	1171 (75.3)	299 (73.3)	599 (74.7)	68 (81.9)	205 (78.2)
*Work status, n (%) (participants could select multiple work status)						
	Retired	1084 (69.7)	285 (69.9)	542 (67.6)	60 (72.3)	197 (75.2)
	Working (part-time or full-time)	125 (8.0)	37 (9.1)	57 (7.1)	11 (13.5)	20 (7.6)
	Volunteer	75 (4.8)	7 (1.7)	27 (3.4)	11 (13.5)	30 (11.5)
	Other	37 (2.4)	9 (2.2)	16 (2.0)	2 (2.4)	14 (5.3)
Household income, n (%)	<$C20 000	86 (5.5)	3 (0.7)	5 (0.6)	21 (25.3)	57 (21.8)
	$C20 000–49 999	331 (21.3)	57 (14.0)	189 (23.6)	15 (18.1)	70 (26.7)
	$C50 000–$99 999	654 (42.1)	184 (45.1)	370 (46.1)	20 (24.1)	80 (30.5)
	≥$C100 000	399 (25.7)	147 (36.0)	178 (22.2)	26 (31.3)	48 (18.3)
	Do not know/prefer not to answer	85 (5.5)	17 (4.2)	60 (7.5)	1 (1.2)	7 (2.7)
Smoking status, n (%)	Current smoker	31 (2.0)	11 (2.7)	16 (2.0)	1 (1.2)	3 (1.1)
	Former smoker	594 (38.2)	195 (47.8)	298 (37.2)	36 (43.4)	65 (24.8)
	Never smoked	930 (59.8)	202 (49.5)	488 (60.8)	46 (55.4)	194 (74.1)
Alcohol consumption past year, n (%)	Regular drinker (> once a month)	1041 (83.3)	315 (88.0)	552 (81.5)	42 (87.5)	132 (79.0)
	Occasional drinker (< once a month)	206 (16.5)	42 (11.7)	123 (18.2)	6 (12.5)	35 (21.0)
	Non-drinker	3 (0.2)	1 (0.3)	2 (0.3)	0 (0.0)	0 (0.0)
Comorbidities, n (%)	None	54 (3.5)	21 (5.1)	21 (2.6)	2 (2.4)	10 (3.8)
	<3	522 (33.6)	126 (30.9)	270 (33.7)	30 (36.1)	96 (36.7)
	≥3	979 (62.9)	261 (64.0)	511 (63.7)	51 (61.5)	156 (59.5)

Continued

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Table 2 Continued

		Total	Hamilton site		Dixon Hall site	
			Males n (%)	Females n (%)	Males n (%)	Females n (%)
Prescribed medication, n (%)	Yes	1261 (81.2)	355 (87.5)	652 (81.4)	71 (85.5)	183 (69.8)
	No	291 (18.8)	51 (12.5)	149 (18.6)	12 (14.5)	79 (30.2)

*Data on working status was added later to the protocol, and data collection for this variable is still ongoing.
BMI, body mass index; MacM3, McMaster Monitoring My Mobility; Standard deviation, SD.

(62.4%) were married or living with a partner, and 75.3% had attained a post-secondary diploma. Additionally, 59.8% reported never having smoked, and 83.3% were regular drinkers. In terms of health-related measures, 62.9% reported having three or more comorbidities, with cataracts, osteoarthritis and hypertension being the most reported conditions. Furthermore, 81.2% were taking prescribed medication(s), and 44.2% and 41.6% rated their general health and mental health, respectively, as 'very good.' Further details of the cohort are provided in online supplemental 1.

More than two-thirds of the participants were born in Canada (69.4%). Most identified as white/Caucasian (88.0%), with smaller proportions identifying as Chinese (7.0%) or another ethnic background such as black, Japanese, South Asian, Latin American, North American Indigenous, Métis, Arab or Korean (5.7%). Ethnic diversity was more pronounced at the Dixon Hall site, where less than half of the participants were born in Canada. In terms of cultural background, participants frequently reported English (40.0%), Scottish (25.1%), Irish (24.1%) and Canadian (14.5%) heritage, with others identifying German, Chinese, French, Italian, Polish, Dutch or other cultural backgrounds. Cultural diversity was also more evident at the Dixon Hall site, where 38.6% of male participants and 27.1% of females reported Chinese cultural heritage, and a wide range of other cultural identities were selected, including Ukrainian, Métis and North American Indigenous backgrounds (online supplemental 1).

Findings to date

Table 3 describes the self-reported mobility and performance-based test results at baseline. Most participants (93.8%) did not use mobility aids, and 37.6% reported experiencing a fall in the previous year. According to the PCML classification, 56.7% of participants had no mobility limitations, while 15.9% had preclinical mobility limitations, and 27.4% had minor mobility limitations at baseline.

For the performance-based tests, the mean of the best right-hand or left-hand grip strength in the overall cohort was 27.3 kg (SD=9.4). The mean gait speed was 1.23 m/s (SD=0.02), the mean TUG test time was 9.4 s (SD=2.5) and the mean 5× STS test time was 13.0 s (SD=3.8). Results are also stratified by sex. Males generally demonstrated higher grip strength and stair climbing power, while other functional measures were relatively consistent across groups. Overall, these results reflect a relatively high-functioning sample of older adults at baseline, which is consistent with our recruitment strategy.[94]

Preliminary results: wearables data

At baseline, data from 1412 participants were processed. Wear time in our study was defined as at least 10 hours per day on a minimum of 7 days. Based on this criterion, 62 participants had no valid wrist wear data, and 49 wore the wrist device for fewer than seven valid days. A total of 1301 participants had valid wrist-worn device data, with an average wear time of 13.6 hours per day (range: 7.2–18.3 hours) over 10.2 days. For the thigh-worn device, 309 participants declined to wear it at baseline, and 95 did not meet the wear-time criteria. As a result, data from 1008 participants are presented. On average, participants wore the thigh device for 13.6 hours per day (range: 6.1 to 22.5 hours) over approximately 10.6 days.

Table 4 presents the mean and median daily step counts by quartile. Step counts ranged from 5073 steps in the lowest quartile to 12 303 steps in the highest. Participants with higher activity levels (Q3: 72.9 years; Q4: 72.7 years) were slightly younger than those with lower activity (Q1: 74.6 years; Q2: 74.7 years). Mobility status also varied with activity level: 73.8% of those in the highest quartile reported no mobility limitations, versus only 36.9% in the lowest quartile. Wear-time compliance was consistent across all quartiles, with a median of 10 compliant days and average daily wear time ranging from 13.8 to 14.6 hours. These data are consistent with findings from other large cohort studies showing a decline in step counts with advancing age and increasing mobility limitations.[95 96]

Trip data from 1422 participants were processed using the algorithms proposed by Fillekes *et al*,[27 49] as described in table 5. Applying the study's compliance criteria of≥7 days of wrist wear with≥10 hours per day, data from 1323 participants are presented. On average, participants wore the wrist device for 12.1 (SD=1.3) hours. In terms of mobility outcomes, participants recorded an average of 2.0 active trips per day (SD=1.3) and 1.8 passive trips per day (SD=1.2). Active and passive trips ranged from fewer than one trip in the lowest quartile to over three trips in the highest quartiles. Participants in the highest

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Table 3 Baseline self-reported mobility, falls and performance-based tests results

	Total sample n (%)	Males n (%)	Females n (%)
Self-reported mobility, n	1555	491	1064
Preclinical Mobility Disability Scale			
No mobility limitation	881 (56.7)	298 (60.7)	583 (54.8)
Preclinical mobility limitation	248 (15.9)	67 (13.6)	181 (17.0)
Minor mobility limitation	426 (27.4)	126 (25.7)	300 (28.2)
Mobility aids, n	1553	490	1063
Yes	96 (6.2)	30 (6.1)	66 (6.2)
No	1457 (93.8)	460 (93.9)	997 (93.8)
Falls in the past year, n	1555	491	1064
Yes	584 (37.6)	169 (34.4)	415 (39.0)
No	971 (62.4)	322 (65.6)	649 (61.0)
Performance-based tests			
*Grip strength (kg), n	1504	479	1025
Mean±SD	27.3 ± 9.4	37.2 ± 8.3	22.7 ± 5.5
Gait speed (m/s), n	1506	480	1026
Mean±SD	1.23 ± 0.2	1.24 ± 0.2	1.22 ± 0.2
Timed Up and Go (s), n	1505	479	1026
Mean±SD	9.4 ± 2.5	9.5 ± 2.5	9.4 ± 2.6
5× sit-to-stand (s), n	1472	466	1006
Mean±SD	13.0 ± 3.8	12.9 ± 3.5	13.0 ± 4 .0
*Single leg stance (s), n	1467	463	1004
Mean±SD	29.6 ± 22.7	26.4 ± 22.1	31.1 ± 22.8
†400-metres walk test (min), n	1109	381	728
Mean±SD	4.9 ± 0.9	4.8 ± 1.0	4.9 ± 0.9
†Stair climbing power (W), n	1140	389	751
Mean±SD	258.1± 84.0	313.3 ± 89.9	229.5 ± 64.2

seconds, s; kilograms, kg; meters per second, m/s; minutes, min; standard deviation, SD; watt (W)
*Best of left and right test.
†The 400 meters walk test and the stair climbing power test are not conducted at DH site.
DH, Dixon Hall; W, watts.

quartile were, on average, younger and had fewer mobility limitations.

Strengths and limitations

To our knowledge, MacM3 is the *first longitudinal study designed to understand trajectories of mobility decline in a generalisable sample of community-dwelling older adults using a body-worn monitoring system that integrates GNSS and accelerometry*. The main strengths of the MacM3 study include its repeated measures and extended follow-up period, its innovative approach to mobility monitoring with advanced wearable technologies, the richness of the health and mobility data collected and the cohort's high participation and retention rate to date.

Although the use of wearable technology to measure real-world mobility holds great promise for improving our understanding of healthy ageing, there are several challenges in this field. There is wide variability in the protocols used and parameters reported from wearable technology, with few studies evaluating their validity.[24 97 98] Wear location of a sensor, for example, is critical to generating valid and sensitive outcomes.[99] Additionally, much of the literature on wearable technology to date is from small cross-sectional studies, specific disease groups or single sensor designs that can limit accuracy, generalisability and precision in the study of mobility in older adults.[100] In recognition of some of these issues, a large European consortium—Mobilise-D—was formed to validate digital mobility measures for health and regulatory agencies. Mobilise-D is conducting a series of clinical validation studies in four specific disease cohorts (each n=600) focusing on accelerometry-derived mobility measures, such as gait speed.[24] Our work complements that of Mobilise-D, as MacM3 will be the first cohort dedicated to evaluating digital mobility measures in a large

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Table 4 Baseline characteristics of participants by quartiles of mean steps per day derived from the thigh devices (n=1008)

Characteristics	Total sample	Quartiles of mean steps per day, mean (SD)			
		Quartile 1 (lowest)	Quartile 2	Quartile 3	Quartile 4 (highest)
Total participants, n	1008	252	252	252	252
Age (years), mean (SD)	73.7 (5.4)	74.6 (5.5)	74.7 (5.5)	72.9 (5.2)	72.7 (4.9)
Mobility score, n (%)					
No mob limitation	582 (57.7)	93 (36.9)	149 (59.1)	154 (61.1)	186 (73.8)
Preclinical	166 (16.5)	49 (19.4)	41 (16.3)	47 (18.7)	29 (11.5)
Minor mob limitation	260 (25.8)	110 (43.7)	62 (24.6)	51 (20.2)	37 (14.7)
No. of days with wear (≥7 days)					
Mean (SD)	9.9 (1.2)	9.7 (1.2)	9.8 (1.0)	10.0 (1.5)	10.0 (1.0)
Median	10	10	10	10	10
Hours of wear on compliant days (≥10 hours)					
Mean (SD)	14.1 (1.4)	13.8 (1.4)	14.1 (1.3)	14.1 (1.3)	14.6 (1.2)
Median	14.1	13.7	14	14.1	14.6
Steps per day					
Mean (SD)	8437 (2943)	5073 (962)	7306 (490)	9065 (560)	12 303 (2283)
Median	8175	5278	7318	8971	11 694

mobility, mob; standard deviation, SD
mob, mobility; SD, Standard deviation.

sample of community-dwelling older adults. Additionally, based on our extensive pilot testing,[27] we have innovated a novel multisensor, multinodal approach that combines GNSS and accelerometry to optimise the potential for identifying early changes in mobility that could inform prevention and early intervention strategies from health, environmental and social perspectives. This is a significant step forward in the study of real-world mobility in community-dwelling older adults, as GPS technology can provide an understanding of spatial properties of mobility patterns and environmental influences but lacks the motion and temporal fidelity to provide details of movement or walking. In contrast, accelerometry data provides details of movement and walking kinematics with continuous millisecond resolution but lacks spatial context. By temporally aligning accelerometry and GPS data, we leverage their complementary strengths to enhance the collection and interpretation of comprehensive mobility data, particularly for understanding environment-movement interactions. Furthermore, our cohort's size, diversity (race, ethnicity, income and ability) and range of health conditions and mobility levels will allow us to validate digital mobility measures with broad applicability for early risk assessment and intervention in community-dwelling older adults.

The main limitations of MacM3 include our convenience sample with recruitment from only two sites in Ontario, Canada, with a large proportion of individuals with post-secondary education and higher income levels. Our sample also consists of more females than males. The gender imbalance may similarly affect the generalisability of the findings, as males and females may have different patterns of mobility, health outcomes and health utilisation. Another limitation is that wearing the thigh device was optional in this study, leading to the absence of key gait and posture-related behaviour data for participants who chose not to wear it. Nevertheless, this limitation is mitigated by the high percentage of participants who agreed to wear the device, maintaining the overall usefulness of the dataset for studying gait and posture.

Future plans

Our future research will focus on leveraging the rich, longitudinal, multisensor data from MacM3 to develop predictive models of mobility decline. These models will enable early identification of individuals at risk of developing mobility limitations, guiding targeted interventions to prevent or mitigate disability. A key goal is to explore the integration of wearable-derived mobility metrics with clinical and environmental factors to personalise strategies for promoting functional independence.

Opportunities for extending the follow-up in MacM3 will also be sought to enhance the usefulness of the data platform and allow for rigorous validation of mobility measures obtained through multisensor wearable technology for monitoring and predicting long-term health outcomes in older adults.

Another critical avenue involves advancing the analytical capabilities of our open-source data processing pipeline, enhancing the fusion of GPS and IMU data to investigate

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Table 5 Baseline characteristics of trip mode derived from the wrist device (n=1323)

Characteristics	Total sample	Quartiles of active trip mode, mean (SD)			
		Quartile 1 (lowest)	Quartile 2	Quartile 3	Quartile 4 (highest)
Total participants, n	1323	349	341	318	315
Age (years), mean (SD)	73.8 (5.3)	74.9 (5.6)	73.0 (5.3)	73.7 (4.9)	73.5 (5.4)
Mobility score, n (%)					
No mob limitation	772 (58.4)	171 (49.0)	211 (61.9)	190 (59.7)	200 (63.5)
Preclinical	209 (15.8)	52 (14.9)	49 (14.4)	60 (18.9)	48 (15.2)
Minor mob limitation	342 (25.8)	126 (36.1)	81 (23.7)	68 (21.4)	67 (21.3)
No. of days with wear (≥7 days)					
Mean (SD)	9.8 (4.5)	9.9 (1.6)	10.0 (1.7)	9.8 (1.3)	9.7 (1.1)
Median	10	10	10	10	10
H/d of wear on compliant days (≥ 10 hour)					
Mean (SD)	12.1 (1.3)	11.9 (1.3)	12.0 (1.4)	12.1 (1.3)	12.3 (1.4)
Median	12	11.8	11.9	12.1	12.3
Active trips per day					
Mean (SD)	2.0 (1.3)	0.5 (0.3)	1.5 (0.3)	2.4 (0.3)	3.7 (0.8)
Median	1.9	0.6	1.5	2.4	3.6
		Quartiles of passive trip mode, mean (SD)			
		Quartile 1 (lowest)	Quartile 2	Quartile 3	Quartile 4 (highest)
Total participants, n	1323	336	359	313	315
Age (years), mean (SD)	73.8 (5.3)	74.7 (5.7)	74.4 (5.6)	73.2 (4.6)	72.7 (5.1)
Mobility score, n (%)					
No mob limitation	772 (58.4)	185 (55.0)	193 (53.8)	191 (61.0)	203 (64.4)
Preclinical	209 (15.8)	54 (16.1)	56 (15.6)	53 (17.0)	46 (14.6)
Minor mob limitation	342 (25.8)	97 (28.9)	110 (30.6)	69 (22.0)	66 (21.0)
No. of days with wear (≥7 days)					
Mean (SD)	9.8 (4.5)	9.9 (1.4)	9.8 (1.5)	9.8 (1.2)	9.9 (1.7)
Median	10	10	10	10	10
H/d of wear on compliant days (≥ 10 hour)					
Mean (SD)	12.1 (1.3)	11.8 (1.4)	12.1 (1.3)	12.2 (1.3)	12.2 (1.3)
Median	12	11.5	12	12.1	12.2
Passive trips per day					
Mean (SD)	1.8 (1.2)	0.3 (0.3)	1.4 (0.3)	2.2 (0.2)	3.5 (0.7)
Median	1.8	0.3	1.4	2.2	3.3

H/d, hours per day; mob, mobility; SD, standard deviation.

the interaction between mobility behaviours and environmental contexts. By incorporating machine learning techniques and stakeholder input, we aim to translate these innovations into actionable tools for clinicians, caregivers and older adults themselves. Ultimately, these efforts will contribute to developing scalable solutions that support ageing-in-place, improve health outcomes and enhance quality of life for ageing populations.

Collaboration

We encourage collaboration on projects that align with the goals of our study, particularly those focused on mobility and healthy ageing. The dataset includes wearable sensor data (accelerometry and GPS), questionnaire responses, demographic and health information, as well as data from physical assessments. Researchers interested in accessing the data will be required to submit a proposal

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outlining their intended use, which will be reviewed by the study team. Data will be shared in a de-identified format whenever possible. However, certain restrictions may apply—particularly for GPS data—due to the sensitive nature of location information. Access to such data will require additional justification and safeguards to protect participant confidentiality.

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